

CoRoT-2b

R-1494

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_150806 · created 2026-08-01T14:06:33+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457240.7216 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.221 +/- 0.066
Transit depth (Rp/Rs)^2	4.88 +/- 2.92 [%]
Orbital Inclination (inc)	89.79 +/- 2.06
Airmass coefficient 1 (a1)	0.139 +/- 0.088
Airmass coefficient 2 (a2)	1.48 +/- 1.59
Scatter in the residuals of the lightcurve fit is	55.35 %
Best Comparison Star	None
Optimal Aperture	1.34
Optimal Annulus	6.25
Transit Duration (day)	0.0994 +/- 0.0064

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.221 ± 0.066 vs 0.1667 (deviation 0.82σ)
Duration fit vs published	0.0994 d vs 0.095 d (deviation 0.69σ)
Mid-time O–C	1.9 min
Depth SNR	3.3
Residual scatter	55.35 %

Light curve

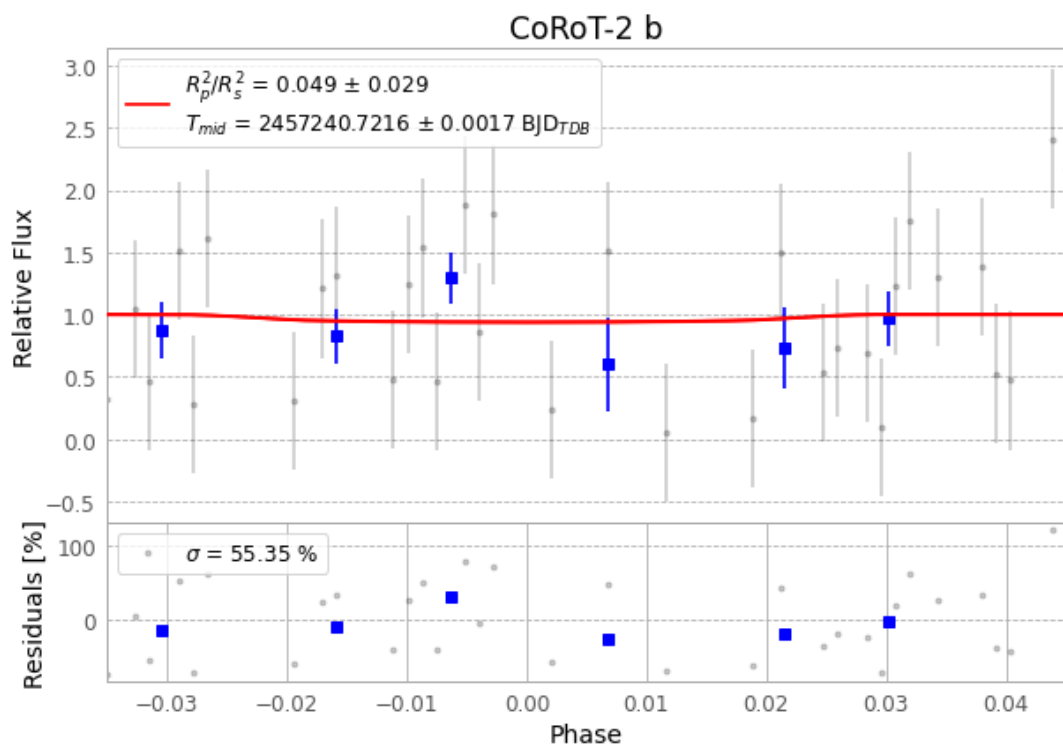


Figure 1 — FinalLightCurve_CoRoT-2 b_2015-08-05.png (SHA-256 4cfe529f01346337...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [318, 284], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e767240c80bd27db...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

72 FITS frames (47 MB), CoRoT-2150806034212.FITS → CoRoT-2150806071512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-150806.log (4d480607eb11...), FinalLightCurve_CoRoT-2 b_2015-08-05.png (4cfe529f0134...), FinalLightCurve_CoRoT-2 b_2015-08-05.csv (1b8cf8decf57...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 14:06Z.